

YEKATERINBURG KOLTSOVO, Russian Federation

WMO: 284400

Lat: 56.743N

Lon: 60.803E

Elev: 233

StdP: 98.56

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.0	-27.8	-34.5	0.1	-30.8	-31.0	0.2	-27.5	9.4	-9.1	8.5	-9.2	2.0	0	0.643

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.9	29.9	19.7	27.9	18.8	26.1	17.9	21.3	26.9	20.2	25.8	19.2	24.5	3.4	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.2	14.4	23.4	18.2	13.5	22.4	17.1	12.6	21.3	63.0	26.8	59.1	25.8	55.4	24.2	24.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
9.2	8.1	7.2		-34.9	32.7	4.0	2.0	-37.8	34.1	-40.2	35.2	-42.4	36.3	-45.4	37.8	
				DB												
				WB	-35.0	22.7	4.0	1.3	-37.9	23.6	-40.3	24.4	-42.5	25.1	-45.4	26.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	2.6	-14.2	-12.4	-4.6	4.0	11.2	16.2	18.1	15.8	10.0	3.2	-5.7	-11.6
	DBStd	12.70	7.12	6.97	5.10	5.24	5.05	4.44	3.71	4.17	4.43	4.86	7.17	7.27
	HDD10.0	3489	750	628	453	189	48	6	1	3	53	215	472	671
	HDD18.3	5851	1008	861	711	429	226	91	51	102	252	468	722	929
	CDD10.0	781	0	0	0	9	86	191	251	184	54	6	0	0
	CDD18.3	101	0	0	0	0	4	27	43	24	2	0	0	0
	CDH23.3	1155	0	0	0	4	98	291	465	269	29	0	0	0
	CDH26.7	307	0	0	0	1	14	73	128	86	5	0	0	0
Wind	WSAvg	3.2	3.1	3.1	3.6	3.6	3.4	3.2	2.8	2.8	3.0	3.5	3.5	3.3
Precipitation	PrecAvg	548	27	19	25	32	51	75	91	72	47	44	36	29
	PrecMax	712	77	39	61	92	102	136	195	150	91	93	80	61
	PrecMin	373	6	2	4	2	8	29	37	22	8	4	5	7
	PrecStd	90	16	10	16	24	24	31	46	34	20	24	18	13
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.9	3.1	10.0	22.9	28.2	31.2	32.2	32.2	26.9	18.9	8.9	2.4
		MCWB	0.5	1.1	5.0	12.4	15.8	19.6	20.0	20.1	16.0	12.0	6.5	1.3
	2%	DB	-0.5	1.1	6.2	18.9	26.1	28.9	29.9	29.1	23.1	15.6	6.0	1.1
		MCWB	-1.6	-0.6	2.6	10.0	15.0	19.0	20.0	19.7	14.9	10.0	4.9	-0.2
	5%	DB	-2.1	-0.5	4.0	15.7	23.9	26.8	28.0	26.2	19.9	12.9	4.0	-0.1
		MCWB	-3.1	-2.1	1.2	8.5	14.3	18.2	19.6	19.0	14.0	8.6	2.8	-1.2
	10%	DB	-4.0	-2.1	2.3	12.9	21.1	24.6	26.0	23.8	17.2	10.2	2.2	-1.9
		MCWB	-5.0	-3.5	0.1	7.2	12.8	17.3	18.7	18.0	12.7	7.6	1.0	-3.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.6	1.5	5.2	13.0	18.5	22.6	22.5	22.4	17.9	12.6	6.9	1.5
		MCDB	1.7	3.1	8.6	21.2	24.1	27.2	27.4	27.3	22.6	17.2	8.2	2.3
	2%	WB	-1.6	-0.5	2.9	10.8	16.3	20.5	21.4	21.0	16.0	10.5	4.8	0.0
		MCDB	-0.7	0.9	5.6	16.4	22.9	26.2	27.1	26.8	20.9	14.2	5.6	0.8
	5%	WB	-3.2	-1.9	1.5	9.2	14.9	19.2	20.4	19.7	14.5	9.1	2.8	-1.2
		MCDB	-2.3	-0.6	3.4	14.7	21.9	24.8	26.0	25.0	18.8	12.2	3.7	-0.3
	10%	WB	-4.9	-3.6	0.4	7.6	13.6	18.0	19.4	18.3	13.3	7.7	1.2	-2.9
		MCDB	-4.0	-2.3	2.2	12.3	20.0	23.3	24.8	22.9	16.6	10.0	2.0	-2.0
Mean Daily Temperature Range	5% DB	MDBR	8.3	10.2	10.0	10.5	12.3	11.0	10.9	10.3	9.4	6.8	6.3	7.3
		MCDBR	6.4	9.8	11.3	15.5	16.6	14.5	14.9	15.0	13.8	12.2	7.1	5.8
		MCWBR	6.0	8.3	8.3	8.7	8.2	7.1	6.8	7.3	7.4	7.6	6.1	5.4
	5% WB	MCDBR	6.4	8.5	9.3	14.0	14.5	13.0	13.0	13.4	12.4	10.9	6.0	5.7
		MCWBR	6.2	7.5	7.2	8.5	8.2	7.2	6.7	7.3	7.7	7.4	5.6	5.6
Clear-Sky Solar Irradiance	taub	0.281	0.312	0.365	0.410	0.409	0.404	0.435	0.445	0.383	0.350	0.307	0.278	
	taud	2.066	2.060	1.973	2.096	2.232	2.317	2.222	2.166	2.308	2.263	2.145	2.025	
	Ebn at Noon	570	713	763	796	827	838	798	756	757	672	535	452	
	Edh at Noon	60	92	131	137	127	118	128	127	95	76	56	48	
All-Sky Solar Radiation	RadAvg	0.69	1.67	3.09	4.10	5.01	5.45	5.30	4.05	2.59	1.27	0.60	0.41	
	RadStd	0.07	0.14	0.24	0.49	0.52	0.49	0.42	0.41	0.20	0.25	0.07	0.07	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	+0.71	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page